

Without ICEPOT

You're right — the request “Generate test cases for file upload” is incomplete. To make it actionable, we need to define **file types, size limits, success/failure criteria, and security considerations**. Let me show you how a complete set of test cases might look once those details are included:

✓ Functional Test Cases

- **Valid file types:** Upload .jpg, .png, .pdf to confirm acceptance.
- **Invalid file types:** Try .exe, .bat, .sh to ensure rejection.
- **File size limits:** Upload files just below, at, and above the max size (e.g., 5 MB).
- **Multiple uploads:** Upload several files simultaneously to check handling.
- **Empty file:** Upload a 0-byte file to see if it's rejected.

⚠ Negative & Error Handling

- **Corrupted file:** Upload a partially corrupted file to test error messages.
- **Network interruption:** Simulate disconnect during upload.
- **Duplicate file:** Upload the same file twice to check duplicate handling.
- **Timeout handling:** Upload a large file with slow connection to test timeout.

🔒 Security Test Cases

- **Malicious file upload:** Attempt to upload scripts disguised as .jpg or .pdf.
- **Path traversal:** Try filenames like ../../etc/passwd.
- **XSS in filename:** Upload a file named <script>alert(1)</script>.jpg.
- **Virus scanning:** Ensure antivirus or scanning mechanism blocks infected files.

🎯 Success Criteria

- **Successful upload:** File appears in the system with correct metadata.
- **Error messages:** Clear, user-friendly error messages for invalid cases.
- **UI feedback:** Progress bar, confirmation message, or preview shown.

This way, the test suite covers **functionality, robustness, and security**.

Would you like me to expand this into a **step-by-step test case table** (with input, expected output, and pass/fail criteria), or keep it as a **high-level checklist** for now?

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Here's Phase 1 — the clarification stage before we design the full enterprise-level test suite.

Clarification Questions

1. **Application Domain** Which domain does the File Upload feature belong to — E-commerce, Banking, Healthcare, Insurance, CRM, HRMS, Document Management System, or Learning Management System? The domain strongly influences file types, security, compliance, and performance requirements.
2. **Supported File Types** Which file formats are allowed (e.g., PDF, DOCX, XLSX, CSV, TXT, JPG, PNG, GIF, ZIP, MP4)? Are there any explicitly **blocked** formats (e.g., EXE, DLL, BAT)?
3. **File Size Constraints**
 - Minimum and maximum file size supported?
 - Should zero-byte files be accepted or rejected?
 - What is the maximum cumulative size for multiple uploads?
4. **Upload Methods** Which methods are supported:
 - Single file upload
 - Multiple file upload
 - Drag-and-drop
 - Browse file
 - File preview
 - Replace/Delete uploaded file
 - Upload progress indicator
 - Resume upload
 - Download uploaded file
5. **File Name Rules**
 - Maximum length of file names?
 - Allowed/disallowed characters (special characters, Unicode, emoji)?
 - Handling of duplicate file names?
6. **Error Handling & Messaging**

- Should error messages be generic or detailed (e.g., “File type not supported” vs. “Upload failed”)?
- Should success/failure messages be localized?

7. Security Requirements

- Should the system block malware/executable uploads?
- Should it validate MIME type vs. extension?
- Should it prevent SQL Injection/XSS via filename?
- Should it handle double extensions (e.g., file.jpg.exe)?
- Should path traversal attacks (../../etc/passwd) be tested?

8. Performance Requirements

- Expected maximum file size for performance testing?
- Expected number of concurrent users uploading files?
- SLA for upload completion time?

9. Accessibility Requirements

- Should the upload feature comply with WCAG 2.1 AA?
- Should screen readers announce progress/error messages?
- Should drag-and-drop be keyboard accessible?

10. Cross-Platform Requirements

- Which browsers (Chrome, Firefox, Safari, Edge, IE11 if legacy)?
- Which devices (desktop, tablet, mobile)?
- Should mobile responsiveness be validated?

11. API Support

- Is there a backend API for file upload?
- Should API validation be included in the test suite?

12. Storage & Session Handling

- What happens if storage quota is exceeded?
- What happens if session times out during upload?
- Should interrupted uploads be resumable?

⚠ Missing Requirements

- Application domain (critical for tailoring test cases).
- Supported/unsupported file types.
- File size limits (min, max, cumulative).
- Upload methods supported.
- File name rules (length, characters, duplicates).
- Error message standards (generic vs. detailed, localization).
- Security policies (malware, spoofing, injection, traversal).
- Performance SLAs (file size, concurrency, response time).
- Accessibility compliance level.
- Cross-browser/device scope.
- API availability.
- Storage/session handling rules.

Once you provide these clarifications, I'll generate the **complete enterprise-level test suite** with all categories (functional, negative, BVA, EP, exploratory, regression, smoke, security, performance, accessibility, automation feasibility).

Would you like to start by confirming the **application domain** first, since that drives most of the other requirements?